

CS6630 : Assignment 3

Fault Attack : 100 Marks

Deadline: April 2nd

Contact for doubts: cs20d408@smail.iitm.ac.in

Benji has access to the device where an encryption algorithm is running. Benji used fault injection techniques to obtain some correct and faulty ciphertext pairs. You have been given information about the online phase of the attack. Your mission, should you decide to accept it, is to perform the offline phase of the attack.

Let's use the following assumptions to find the secret key.

Assumptions:

- Fault Model Used: Single-Byte Fault
- Cipher Algorithm Used for Encryption: AES-128 ECB
- Drive link:  CS6630_Fault_assignment

To be Answered:

1. Identify the round at which the fault was injected. Justify your answer using the ΔC (XOR difference) patterns observed from the ciphertext pairs. (10 marks)
2. Which round key were you able to obtain directly from the DFA offline analysis? (5 marks)
3. The number of key bytes retrieved after the offline analysis? (5 marks)
4. Python script to perform the offline analysis (55 marks)
Script must:
 - a. Read correct and faulty ciphertexts
 - b. Identify the fault round automatically
 - c. Recover the round key via DFA
 - d. Reverse the key schedule to obtain K0
 - e. Verify: `AES(plaintext, K0) == correct ciphertext`
 - f. Output all results to roundkeys.txt
5. Detailed Report (25 marks)
 - a. Describe the steps used for the Analysis
 - b. Time taken to obtain the round key?
 - c. Minimum pair of ciphertext/faulty ciphertext pair to retrieve all bytes of round key?
 - d. List of LLM prompts you have used in detail.

Submission Format:

Folder rollnum should contain the following files

1. Python script for the analysis - **dfa_8.py**
2. **roundkeys.txt** should include the round keys values obtained

Example:

round 1 : [117, 4, 184, 16, 195, 49, 154, 87, 75, 44, 133, 255, 97, 59, 253, 136]

3. Report (in PDF)

PS: Evaluation will be strictly based on this format. Please make sure you follow the format for the first two files.